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Reconfigurations

Agency is not an attribute but the ongoing reconfigurings of the world.
(Barad 2003: 818)

In this chapter I consider some new resources for thinking about, and acting within, the interface of persons and things. It is here, on the question of alternatives to information theoretic approaches to human–machine interactions, that I believe the ground has shifted most radically over the past twenty years. The shifts involve reconceptualizations of the social and the material and the boundary between them, with associated implications for practices of system design. The explorations are ongoing within relevant areas of cultural anthropology, science and technology studies, feminist theory, new media studies, and experiments in cooperative systems design, each of which is multiple and extensive in themselves and no one of which I can do full justice to here. I hope nonetheless to trace out enough of the lines of resonance that run through these fields of research and scholarship to indicate the fertility of the ground, specifically with respect to rethinking and creatively enacting the interface of humans and machines.

EXCLUDED MIDDLES

It was the circular move of writing a cognitivist rationality onto machines and then claiming their status as models for the human that first provoked me to question the notion of intelligent, interactive artifacts. My concern then, as now, has to do with the implications of this move both for our notion of what machines are and also with ways in which the premises and products of artificial intelligence research continue to restage traditional Euro-American assumptions about the nature of being human. In setting up my critique, however, I fell back into a familiar humanist stance, defending against what I saw as AI's attributions of (a certain version of) intelligence and interactivity to machines by effectively reclaiming (a different version of) those qualities for humans. Since then, I have struggled with the question of how to maintain the sense of human–machine difference that I developed in my analysis, while taking to heart the insights generated from subsequent thinking regarding the distributed and enacted character of agency, and the implications of such reconceptualizations for essentialist human–nonhuman divides.

Latour (1993: 77–8) usefully demarcates a "Middle Kingdom" with respect to human–nonhuman relations, within which he locates the space between simple translations from human to nonhuman, on the one hand, and a commitment to maintaining the distinctness and purity of those categories, on the other. Translation in the case of humans and machines involves practices through which capacities taken to be inherent in one are shifted to, or realized through, the other. In resisting the particular translations of intelligence and interactivity recommended by AI in the 1970s and 1980s, I turned to a kind of exercise of purification, attempting to maintain those qualities as exclusively human. I now believe that what we need is to, in Latour's words, "direct our attention simultaneously to the work of purification and the work of hybridization" (1993: 11) with respect to human–machine boundaries. This involves developing a discourse that recognizes the deeply mutual constitution of humans and artifacts, and the enacted nature of the boundaries between them, without at the same time losing distinguishing particularities within specific assemblages. Recognizing the interrelations of humans and machines, in other words, does not mean that there are no differences. The problem rather is how to understand the nature of difference differently.[1]

I want to wander about a bit in Latour's Middle Kingdom, then, in considering the question of agency in humans and machines. For those like Latour writing within the Actor Network framework and its

[1] Questions of difference have been most extensively considered within feminist and postcolonial scholarship. For some exemplary texts see Ahmed (1998, 2000); Ahmed et al. (2000); Bhabha (1994); Braidotti (1994, 2002); Castañeda (2002); Franklin, Lury, and Stacey (2000); Gupta and Ferguson (1997); Strathern (1999); Turnbull (2000); Verran (2001).

aftermath, agency is understood as a material-semiotic attribute not locatable in either humans or nonhumans.[2] Agency on this view is rather an effect or outcome, generated through specific configurations of human and nonhuman entities. Moreover, in a move echoing the Melanesian conception of personhood described by Strathern (see Chapter 14), the entities involved do not precede their incorporation into such configurations in any simple way but emerge through their participation in various networks of relations. In the words of Callon, the network of interest for Actor Network Theory (ANT) is "not a network connecting entities which are already there, but a network which configures ontologies. The agents, their dimensions, and what they are and do, all depend on the morphology of the relations in which they are involved" (1999: 185–6). ANT's call for a "generalized symmetry" in analyses of human and nonhuman contributions to social order performed a powerful intervention into sociological preoccupations with human agency, as the latter "[l]iberated from its containment in human entities . . . is dispersed through the networks" (Ashmore et al. 1994: 2). I return to the question of symmetry below. But I turn first to the rich body of empirical studies that have specified, elaborated, and deepened the senses in which human agency is only understandable once it is reentangled in the sociomaterial relations that the "modern constitution" (Latour 1993) has, since the seventeenth century, so exhaustingly attempted to take apart.

MUTUAL CONSTITUTIONS

A growing corpus of studies of sites of sociomaterial practice over the past twenty years provide compelling empirical demonstrations of how capacities for action can be reconceived on foundations quite different from those of a humanist preoccupation with the individual actor living in a world of separate things. This body of work is too extensive to be comprehensively reviewed, but a few indicative examples can serve as illustration.

[2] The phrase *material-semiotic* was coined by Haraway (1991: 194–5) to indicate the ways in which the natural and the cultural, or the material and the meaningful, are inextricably intertwined. Although not cited in the early formulations of Actor Network Theory, the writings of Haraway and other feminist science studies scholars have since become increasingly central to writings "after" ANT. See, for example, the articles collected in Law and Mol (2002). I return to regenerative discussions of agency and difference within feminist scholarship below.

The question of human–nonhuman relations has been intensively explored within science studies. Pickering (1995) develops the metaphor of the "mangle" to create a performative account of knowledge practices, including centrally the construction of machines that "variously capture, seduce, download, recruit, enroll or materialize" human agency (ibid.: 7). Key to Pickering's analysis is time, the view that what he names *material agency* is always temporally emergent in practice, rather than fixed in either subjects or objects (see also Lynch, Livingston, and Garfinkel (1983)). Knorr-Cetina adopts a trope of "epistemic cultures" to think about laboratories as mutually shaping arrangements of scientists, instruments, objects, and practices aimed at the production of observably stabilized instantiations of "reality effects" (1999: 26–33). The notion of reconfiguration is central to her analysis as well, as the process through which subject/object relations are reworked. Considered over time, she argues, reconfigurations comprise what are commonly termed skills or expertise: "The alignments . . . work through the body of the scientist, but they also involve a drastically rearranged environment, a new life-world in which new agents ineract and move. When we ascribe skills to a person . . . the person acts as a symbol – a stand-in for the common life-world with objects, which, in the laboratory . . . is continually recreated" (219–20). Knorr Cetina's argument here has resonance as well with Lynch's (1991) formulation of "topical contextures" to indicate the inseparability of knowledge practices and the phenomenal fields of action that they at once constitute and inhabit, and with Ingold's (2000) analysis of skill not as an attribute of a body, but of a system of relations involving the artisan's presence in a specifically configured sociomaterial environment.

In his exploration of what he terms "professional vision," Charles Goodwin has carried out a series of studies focused on the sociomaterial interactions through which practitioners learn to see the phenomena that constitute the objects of their profession (C. Goodwin 1994, 1995a, 1997, 2003). A central argument is that these phenomena are not preexisting but are constituted as disciplinarily relevant objects through occasioned performances of competent seeing (see also Goodwin and Goodwin 1996, 1997). In looking at gestures and their objects, for example, Goodwin argues that the relation is a "symbiotic" one; that is, "a whole that is both different from, and greater than its parts, is constructed through the mutual interdependence of unlike elements" (2003: 20). Symbiotic gestures, Goodwin argues, are not referring to something outside of themselves: rather, the gesture's objects are integral components of the gesture itself (ibid.: 40, note 1). In the case of archaeologists "defining features" of relevance in a site of excavation, for example, Goodwin observes that a "feature" does not simply present itself but must be made visible through the embodied work of the archaeologist, including talk with colleagues, gestures, inscriptions in the dirt, and various forms of record keeping, mapping, and the like (see also Latour 1999: 58–61). In this way "a feature as a semiotic object . . . emerges as the product of both actual patterning in the soil being investigated, and the cultural categories and embodied practices used by archaeologists to make it visible as a particular kind of phenomenal object" (ibid.: 29). At the same time, the objects being defined and their categorization exist within a professional matrix of social and material accountability, subject to contest by the readings of others and by the objects themselves; for example, in the discovery of roots extending from what has been previously identified as a post mold, indicating instead the presence of a tree (ibid.: 30). Archaeological knowledge, on this analysis, comprises relations between particular culturally and historically constituted practices and their associated materials and tools. It is out of those relations, quite literally, that the objects of archeological knowledge and the identity of competent archeologist are co-constructed.

Although not concerned specifically with interactive machines, Goodwin's analysis provides further support for the wider argument against attributions of agency either to humans or to artifacts and gives us, in turn, a different way of understanding the problem of attributions of knowledge and agency to machines. The problem is less that we attribute agency to computational artifacts than that our language for talking about agency, whether for persons or artifacts, presupposes a field of discrete, self-standing entities.[3] As an alternative, we can take the interface not as an a priori or self-evident boundary between bodies and machines but as a relation enacted in particular settings and one, moreover, that shifts over time.

The shifting nature of body–machine boundaries is enacted quite literally in the case of technology-intensive medicine, and here again an instructive series of studies are available. Dawn Goodwin (2004) describes the practices through which patients in surgery are "transitioned" through anaesthetic states, a process involving the radical reconfiguration of their capacity for action; specifically, for the sustenance

3 Latour makes a closely related argument, using the example of the gun. See Latour (1999: 179–80). See also Casper (1994), Law (1987).

of their own life support. Over the course of an anaesthesia, agencies involved in the maintenance of vital bodily functions are progressively delegated from the patient as an autonomously embodied entity to an intricately interconnected sociomaterial assemblage and then back again.[4] Through a series of cases, Goodwin demonstrates how the technologies of anaesthesia are joined to the patient's body, in ways that render the latter highly dependent and vulnerable but nonetheless intensely (albeit sometimes ambiguously) communicative. This joining is analyzed as a delicate choreography involving patients, medical practitioners, and machines.[5] Goodwin argues that questions of agency are crucial both to assess policy with respect to medical practice and to deepen our understanding of the dense sociotechnical arrangements that comprise much of contemporary medical activities and institutions.

In a related argument developed through the case of reproductive technoscience, Thompson (Cussins 1998; Thompson 2005) argues against the idea that medical interventions inherently objectify patients and thereby strip them of their agency. She observes that in the case of infertility clinics "the woman's objectification, naturalization, and bureaucratization involve her active participation and are managed by herself as crucially as by the practitioners, procedures, and instruments" (Cussins 1998: 167). Conversely, objectification does not inherently or necessarily lead to alienation, nor does it stand always in opposition to subjectivity or personhood. Among other things, the clinic relies on the possibility of separation (of egg and sperm from the bodies that produce them) without alienation. Cussins locates alienation not in objectification per se, but in the breakdown of synechdochal relations between parts and whole that make objectification of various forms into associated forms of agency. It is this process "of forging a functional zone of compatibility that maintains referential power between things of different kinds" that she names *ontological choreography* (ibid.: 192). Medical ethics and accountability, she argues, need to be founded

4 The particular expertise of the anaesthetic practitioner on this account is to manage the often unruly contingencies of the unfolding course of anaesthesia through a combination of skillfully embodied techniques, reading of signs, professional judgments, and legitimating accounts, which together provide the grounds for practical action. See also Heath et al. 2002, Hirschauer 1991, Mort et al. 2005.

5 The trope of "choreography" was introduced by Charis Thompson (Cussins 1998, Thompson 2005), whose work I return to below.

not in the figure of the rational, informed citizen but in the conditions for the maintenance of those crucial relations that configure identities and selves and that might allow them to be reconfigured in desired ways.

The assemblage of the pregnant woman has been the focus as well of Casper's research on experimental fetal surgery, where categories of the human and associated agencies take on a particular salience and urgency (1994, 1998). In the context of debates over abortion within the United States, the figuring of "fetal patients" has consequences that resonate not only within but also well beyond the walls of the surgery. In addressing this tricky political terrain, Casper calls for a methodological strategy aimed at "grounding the construction of social identities and subject positions in concrete practices, more specifically the practices through which fetal humanity, including agency, is socially and technologically shaped" (1994: 2). A central moral of Casper's story is that questions of agency are inseparable from the more extended frames of reference in which entities are entangled or, alternatively, that their separation is itself a strategically consequential act. I return to the question of frames below but note for the moment that Casper's analysis suggests in turn that the politics of fetal agency cannot be adequately debated without taking as our primary unit the woman *plus* fetus, *within* the context of the latter's contested material and symbolic status and its implications for actual women's lives.

Whereas fetal surgery would stand as among the most maximally invasive of medical procedures, a different sense of the fluidity of body–machine boundaries is provided by Prentice's ethnographic interviews with physicians engaged in its surgical opposite (2005). Minimally invasive or "keyhole" surgery, as it has developed over the past few decades, has involved a series of shifts in the gaze of the surgeon and attendant practitioners from the interior of the patient's body – formerly achieved through a correspondingly large incision – to views mediated first through microscopy and now through digital cameras and large screen monitors. Prentice finds that surgeons accustomed to operating within previous configurations of patient and instruments express a sense of disorientation when they are translated into the reconfigured sociotechnical network of video camera and monitor. One surgeon with whom Prentice speaks reports an experience not only of his gaze but also of his hands and entire body, effectively leaving the site of the patient's body and "going to work on the monitor" or image instead; a translation that he finds deeply alienating.[6] In contrast, Prentice found that surgeons who have performed minimally invasive surgery mediated by camera and monitor throughout their career report a very different phenomenal shift. Far from being alienated from the patient, they experience themselves as proprioceptively shifted more directly and proximally into the operative site, with the manipulative instruments serving as fully incorporated extensions of their own acting body.[7] As Prentice observes of these cases: "When the patient's body is distributed by technology, the surgeon's body reunites it through the circuit of his or her own body" (ibid.: 8). These differences suggest again that questions of alignment or dislocation, relation or alienation, are not immanent in human–machine boundaries or even, a priori, in particular human–machine configurations. Rather, they are effects lived and experienced within multifaceted subject–object assemblages.

The shifting boundaries of humans and machines and their consequences comprise the topic of another study of minimally invasive surgical practices by Aanestad (2003), who focuses on the labors performed by nurses and technicians in aligning the complex sociotechnical environment of the surgical theatre itself.[8] Her study follows the installation of multimedia communications technologies (cameras, microphones, and speakers) in a surgical operating theatre in ways intended not only to enable the surgery but also to facilitate communication with viewers outside, including with remotely located surgeons in training. Aanestad's analysis follows the course of shifting interdependencies in the surgical assemblage, as changes to existing arrangements necessitate further

6 It is important to note that this is not a simple distinction between mediated and unmediated access. All of the surgeons with whom Prentice spoke in this study were experienced in keyhole or minimally invasive surgery. The surgeon who reported his sense of disorientation with the latest techniques had previously worked while looking through a microscopic eyepiece; the disruptive shift for him was from that to a video monitor more distal from both his own and the patient's body.

7 This sense of the fluidity of body boundaries and their reconfigurability is resonant with Mol's findings (2002) regarding the ontologies of subjects, artifacts, and objects in medical practice.

8 The question of visibility–invisibility and framing resonates throughout Aanestad's study, as nurses and technicians configure the theatre for transmission of the surgery to remote audiences in ways that center the surgeon and quite literally relegate their own work to the margins, outside the field of view. At the same time, this is not a simple story of power lost, as the technologies become available to them for appropriation in new ways, while their own role in the surgical process becomes more indispensable. On invisible work see Clement (1993), Shapin (1989), Star (1991), Suchman and Jordan (1989); on gendered (re-)appropriations of new technologies see Cherny and Weise (1996), Spender (1996), Terry and Calvert (1997), Wakeford (2000), Wolmark (1999).

changes in a process that she names the in situ work of "design in configuration" (2003: 2). She emphasizes that the agencies of the technologies involved do not exist before their incorporation into the network; for example, as questions of the adequacy of image and sound quality or shifts in the locus of control. Aanestad concludes that introducing telemedicine or other network technologies in such settings requires "open and evolutionary strategies, which are aimed at enrolling allies, rather than control-oriented, specification-driven strategies" (ibid.: 16). Her analysis makes clear how in such a setting the capacity for action is relational, dynamic, and collective rather than inherent in specific network elements and how the extension of the network in turn intensifies network dependencies.

Together these inquiries respecify sociomaterial agency from a capacity intrinsic to singular actors to an effect of practices that are multiply distributed and contingently enacted. Addressing similar questions, but from a position within feminist philosophy and science studies, physicist Karen Barad has proposed a form of materialist constructivism that she names "agential realism," through which realities are constructed out of specific apparatuses of sociomaterial "intra-action" (2003). Whereas the construct of interaction suggests two entities, given in advance, that come together and engage in some kind of exchange, *intra-action* underscores the sense in which subjects and objects emerge through their encounters with each other.[9]

More specifically, Barad locates technoscientific practices as critical sites for the emergence of new subjects and objects. Taking physics as a case in point, her project is to work through long-standing divisions between the virtual and the real, while simultaneously coming to grips with the ways in which materialities, as she puts it, "kick back" in response to our intra-actions with them (1998: 112; see also Knorr Cetina (1999), Pickering (1984, 1995), Traweek (1988)). Through her close readings of Niels Bohr, Barad insists that "object" and "agencies of observation" in his view form a nondualistic whole: it is that relational entity that comprises the objective "phenomenon" (1996: 170). In a position consistent with Haraway's adoption of the compound "material-semiotic," Barad takes concepts and their objects as mutually constitutive. Different "apparatuses of observation" enable different, always contingent, subject–object cuts that in turn enable measurement or other forms of

9 Smith (1996) develops a kindred concept of "registration" to describe the partial effects of subject–object difference, generated through processes of engaged participation.

objectification, distinction, manipulation, and the like *within* the phenomenon. The relation is "ontologically primitive" (2003: 815), in other words, or prior to its components; the latter come about only through the "cut" effected through a particular apparatus of observation.

One implication of this view is a more radical understanding of the sense in which "materiality is discursive (i.e., material phenomena are inseparable from the apparatuses of bodily production: matter emerges out of and includes as part of its being the ongoing reconfiguring of boundaries), just as discursive practices are always already material (i.e., they are ongoing material (re)configurings of the world)" (Barad 2003: 822). This intimate co-constitution of configured materialities with configuring agencies clearly implies a very different understanding of the human–machine interface. Read in association with the empirical investigations of complex sociomaterial sites described above, "the interface" becomes the name for a category of contingently enacted cuts occurring always *within* sociomaterial practices, that effect "person" and "machines" as distinct entities, and that in turn enable particular forms of subject–object intra-actions. At the same time, the singularity of "the interface" explodes into a multiplicity of more and less closely aligned, dynamically configured moments of encounter within sociomaterial configurations, objectified as persons and machines. It is the differences effected *within* such configurations that I turn to next.

DIFFERENCES WITHIN

The reconstructions of sociomaterial agency reviewed above are frequently summarized by the proposition that humans and artifacts are *mutually constituted*. This premise of technoscience studies has been tremendously valuable as a corrective to the entrenched Euro-American view of humans and machines as autonomous, integral entities that must somehow be brought back together and made to interact. But at this point I think that the sense of mutual constitution warrants a closer look. In particular, we are now in a position to elaborate that generative trope along at least two critical dimensions: first, in relation to the dynamic and multiple forms of constitution that are evident in specific sociomaterial assemblages and, second, in terms of questions of difference – and more particularly asymmetries – within those assemblages.

As the studies reviewed above and others like them have shown, the constitution of humans and artifacts does not occur in any single time and place, nor does it create fixed human–artifact relations or entities.

Rather, artifacts are produced, reproduced, and transformed through ongoing "labours of division," in Law's phrase (1996), that involve continuous work across particular occasions and multiple sites of use. This work of production and reproduction across time and space results in very diverse assemblages, involving participants with different histories, relations of familiarity or strangeness, and the like. As Mulcahy points out with respect to technologies (1999), it is their increasingly extensive distribution and the range of variations across user–machine pairings that render protocols, standards, instructions, and the like necessary to the successful production and reliable reproduction of human–artifact interactions. Empirical investigations of the workings of standards and other technologies aimed at the reproduction of sameness (including those that take the form of plans or instructions examined earlier in this book) provide ample evidence that the agencies of such artifacts do not inhere in the prescriptions themselves but rely on the skilled practices that bring them into alignment with a given case at hand.

Mutualities, moreover, are not necessarily symmetries. My own analysis suggests that persons and artifacts do not constitute each other *in the same way*.[10] In particular, I would argue that we need a rearticulation of asymmetry, or more impartially perhaps, dissymmetry, that somehow retains the recognition of hybrids, cyborgs, and quasi-objects made visible through technoscience studies, while simultaneously recovering certain subject–object positionings – even orderings – among persons and artifacts and their consequences. The emphasis in science and technology studies on symmetrical analysis and the agency of things arose from well-founded concerns to recover for the social sciences and humanities aspects of the lived world – for example, "facts of nature" and "technology" – previously excluded from consideration as proper sociological subjects. My project is clearly indebted to these efforts, which provide the reconceptualizations needed to move outside the frame of categorical purification and opposition between social and technical, person and artifact. My own engagement with these questions, however, came first in the context of technoscience and engineering, where the situation is in important respects reversed. Far from being excluded, "the

[10] As Pickering points out with respect to humans and nonhumans, "Semiotically, these things can be made equivalent; in practice they are not" (1995: 15). This notwithstanding the possibility of delegating humanlike actions to machines, or identifying machinelike actions within the activities of humans (see also Collins 1990).

technical" in regimes of research and development are centered, whereas "the social" is separated out and relegated to the margins. It is the privileged machine in this context that creates its marginalized human others.[11]

So what are the possibilities for recovering a sense of particular agencies of the human without at the same time reinstating essentialized human–machine differences? How might we reconceptualize the granting of agency in a way that at once locates the particular accountabilities of human actors, while recognizing their inseparability from the sociomaterial networks through which they are constituted? Analyses – including my own – that describe the active role of artifacts in the configuration of networks inevitably seem to imply other actors standing just offstage for whom technologies act as delegates, translators, mediators; that is, human engineers, designers, users, and so on. I want to suggest that the persistent presence of designers–users in technoscientific discourse is more than a recalcitrant residue of humanism: that it reflects a durable dissymmetry among human and nonhuman actors. The response to this observation is not, however, to cry "Aha, it really is the humans after all who are running the show." Rather, we need a story that can tie humans and nonhumans together without erasing the culturally and historically constituted differences among them. Those differences include the fact that, in the case of technological assemblages, persons just are those actants who configure material-semiotic networks, however much we may be simultaneously incorporated into and through them.[12] I want to keep in view as well the ways in which it matters when things travel across the human–artifact boundary, when objects are subjectified (e.g., machines made not actants but actors) and subjects objectified (e.g., practices made methods or knowledges made commodities).[13]

Applied to the question of agency, I have argued that in the case of the intelligent machine we are witnessing a reiteration of traditional humanist notions of agency, at the same time – even through – the

[11] I use "others" here in the sense nicely summarized by Lee and Brown as "all those entities and areas of inquiry that are rendered problematic by expansionist projects, be they formally political or theoretical" (1994: 773).

[12] Pickering (1995: 15) similarly poses the question of who does the "delegation" of agencies across actor networks.

[13] As Haraway (2003: 4) succinctly reminds us with respect to the machinic and animal, "the differences between even the most politically correct cyborg and an ordinary dog matter."

intra-actions of that notion with new computational media. In the remainder of this chapter I look to further experiments in configuring human–machine boundaries to explore the question of what other directions our relations with machines, both conceptually and practically, might take.

REREADING THE HUMAN–MACHINE

I turn first to recent counterreadings of the humanlike machine, inspired by feminist discussions of materialities, subjectivities, and cyborg bodies. Like many, my attention was first drawn to these possibilities by Donna Haraway's "whip lashing" proposal (a phrase that she herself uses to describe those moments when a new idea comes along that turns one's head) that we should all prefer to be cyborgs than goddesses (1985/1991: 223). As Wolmark summarizes, in her discussion of the "Manifesto for Cyborgs": "The cyborg's propensity to disrupt boundaries and explore differently embodied subjectivities could . . . be regarded as its most valuable characteristic, and it is undoubtedly one of the reasons for its continued usefulness in feminist and cultural theory" (1999: 6).[14] As feminist theorists trace a new path across the problematic terrain of how the sexed and gendered subject might be reconceived, they also provide us with resources for reconceptualizing the agential object. More specifically, feminist retheorizing of the body has been concerned to restore the dynamism emptied out of bodies by the mind–body split by moving through that split onto new terrain. In a similar way, feminist theorists suggest that we might find other grounds for recognizing the agential properties of the material than the operations of a transcendental intelligence over inert, mechanistically animated matter. As Butler famously puts it in *Bodies That Matter*: "What I would propose . . . is a return to the notion of matter, not as site or surface, but as a process of materialization that stabilizes over time to produce the effect of boundary, fixity, and surface we call matter . . . Crucially, then, [the construction of bodies] is neither a single act nor a causal process initiated by a subject and culminating in a set of fixed effects"

[14] At the same time, as Balsamo cautions, far from imploding the boundaries of human and machine, for most popular cyborg figures "Signs of human-ness and, alternatively, signs of machine-ness function not only as markers of the 'essences' of the dual natures of the hybrid, but also as signs of the inviolable opposition of human and machine. This is to say that cyborgs embody human characteristics that reinforce the difference between humans and machines" (2000: 149).

(1993: 9–10). Butler's argument that sexed and gendered bodies are materialized over time through the reiteration of norms is suggestive for a view of technology construction as a process of materialization through a reiteration of forms. Butler argues that "sex" is a dynamic materialization of always contested gender norms: similarly, we might understand "things" or objects as materializations of more and less contested, normative figurations of matter. Much as recognition and intelligibility are central to feminist conceptions of the subject, objects achieve recognition within a matrix of historically and culturally constituted familiar, intelligible possibilities. Technologies, like bodies, are both produced and destabilized in the course of these reiterations.

An early example of an alternative cyborgian embodiment is provided by Deirdre, the heroine of science fiction writer C. L. Moore's 1944 short story, "No Woman Born."[15] Deidre prefigures Haraway's challenge to the cultural imaginary of the goddess-turned-cyborg. Ambivalently positioned on the boundary of Cartesian and feminist imaginaries, the premise of Moore's story is that Deirdre, once an exquisitely beautiful and talented dancer, has been injured in a theater fire to the point that only her brain survives. As the brain of a *dancer*, however, Deirdre's brain is located by Moore in intimate relation to her body. As the story unfolds, it becomes clear that the restoration of Deirdre's agency is inseparably tied to the particularities of her rematerialization. We enter the story one year after the tragic fire, during which time Deirdre (Deirdre's brain?) has been painstakingly reembodied by Maltzer, a genius physician–scientist, assisted by a team of unnamed (but apparently greatly talented) sculptors and artists. The story that follows is effectively a set of variations around the theme of Deirdre's rematerialization, haunted by questions of memory, identity, recognition, transformation, and otherness. We approach these questions through the person of John Harris, Deirdre's former (human) agent and close friend, coming to see her for the first time following the accident. Torn by visions of, on one hand, the irrecoverable figure of Deirdre the human as he knew her and, on the other, culturally inspired imaginings of how the new, robotic Deirdre might be configured, Harris suffers agonies of anticipation in advance of their meeting. His anxieties are not allayed by the comments of her restorer Maltzer, in the anteroom of Deirdre's chambers: "It's not that she's – ugly – now . . . Metal isn't ugly. And Deirdre . . . well, you'll see.

[15] It is notable that this story appears in an anthology of science fiction short stories within which C. L. Moore is the only woman author.

I tell you, I can't see myself. I know the whole mechanism so well – it's just mechanics to me. Maybe she's – grotesque, I don't know" (1975: 67).

The Deirdre that Harris goes on to meet is less a replica than a new configuration, a reembodiment, of the Deirdre he remembers. In place of a face, she has a delicately modeled ovoid head with a kind of golden mask, in which a slit of aquamarine crystal occupies the place where her eyes would have been. And rather than a simulation of human skin over hinged metal joints, her body is made up of tiny golden coils, infinitely flexible, covered by a robe of very fine metal mesh, all of which she has learned to move with an extraordinary expressiveness and grace at once reminiscent of, and different from, her former dancer's body. As Harris struggles to come to terms with the neither–nor, both–and qualities of the new Deirdre, the story unfolds as a succession of reflections on the uncertainties of Deirdre's status, in relation to her former identity as Deirdre and to the rest of the human world. First, what is the relation of this new creature to "Deirdre" herself? Is "she" still alive? And what about the reembodied Deirdre's relation to her creator, Maltzer? Is she an extension of him, his property, or an autonomous being, animating the materials that he has provided with her own "unquenchable" essence? And is her essence that of the brain that survived or some irreducible spirit that animates her new body? In one of his more posthumanist moments, Harris muses, "She isn't human, but she isn't pure robot either. She's something somewhere between the two, and I think it's a mistake to try to guess just where, or what the outcome will be" (ibid.: 88). The story's pivotal question, on which the plot turns, is whether Deirdre is still human and, if not, whether the rematerialized Deirdre can survive given her singular otherness. Not surprisingly, this question remains unanswered at the story's end. But what Moore has achieved is to reframe the cyborg from its reiteratively human replicant form to something that dances elusively, and therefore suggestively, on the boundaries of old and new possibilities. Deidre embodies the ambivalences of mid-twentieth-century technoscience, suggesting the possibilities for new configurations that are fabulous and expansive, while at the same time threatening the reassuring ground of normative categories on which our experiences of relationship, of knowing and being known, depend. Figured alternately as goddess, human, superhuman, and monster, Deirdre powerfully expresses the questions raised by new sociomaterial possibilities and their relations to old struggles around identity and difference.

More recently, Claudia Castañeda (2001) has written about the rematerialization of touch in contemporary robotic artificial intelligence.

Beginning from an understanding of touch as always semiotic and relational, and of signs as always entailing materialities, she takes up the question of the skin and its materialization in the form of the robot Cog (see Chapter 13).[16] Interactivity is framed by Cog's designers as the litmus test of its competencies, with "the world" and with its human counterparts. During Cog's early, awkward stage, its "skin" (described as "an exquisitely sensitive piezo-electric membrane" in Dennett 1994: 139) is designed to serve as a protective device against contact, equipped with the requisite sensors and alarms. Castañeda explores the premise that Cog's embodiment, particularly its skin, is designed to change in response to the robot's interactions over time. My skeptical reading of the project falters on the question just how open the possibilities of rematerialization are for Cog given the robot's origins in the historical and cultural matrix of the Massachusetts Institute of Technology. But Castañeda's hopeful reading points us to aspects of Cog that at least signal the possibility of what she names a "feminist robotics" (2001: 233).

First, and most basically, Cog's design (at least on this telling) locates touch as a way of knowing and being in the world. Second, Castañeda suggests that Cog embodies a relational conception of the body, one that extends beyond the boundaries of the skin and that is generated through particular, changing combinations of materials and qualities. And finally, as she puts it, Cog is "neither human nor anti-human, but rather other-than-human" (ibid.: 232). As such, she argues that Cog's reembodiment of the human in different terms generates the possibility, in material form, of embodied alterity, a relation of difference that literally as well as figurally matters. Castañeda's interest, then, is in just what kind of alterity is, or could be, embodied in the robot, which does not take the human, normatively imagined, as the "origin and truth against which the robot's value is always measured" (ibid.: 234).[17]

The question of how the robot could be other than second term to the human aligns with feminist concerns regarding what Anne Balsamo sums as "the systems of differentiation that make the body meaningful,"

[16] It is critical to Castañeda's reading of Cog that she relies on an account of Cog's conception offered by philosopher of mind Daniel Dennett (1994) rather than on accounts or observations of the robot as implemented. This does not diminish the suggestive possibilities of her analysis, only the question of their realization within prevailing robotic imaginaries.

[17] For another reflection on the robot's current and potential figurations, see Castañeda and Suchman (in press).

most notably those of gender (1996: 21). Power works through binary opposites not in the simple sense that the first term holds power over the second but that their relative positionings – including, crucially, as opposites – enable their fundamental interrelatedness and the historically sedimented cuts that position them as separate categories to be obscured. In contrast, Judith Halberstam proposes that in her feminist conceptions: "The intelligent and female cyborg thinks gender, processes power, and converts a binary system of logic into a more intricate network" (1991: 454). Framed not as the importation of mind into matter, but as the rematerialization of bodies and subjectivities in ways that challenge familiar assumptions about the naturalness of normative forms robots, and cyborg figures more generally, become sites for change rather than just for further reiteration.

Feminist rereadings of the cyborg replace the binaries male–female, human–machine, and subject–object with the possibility of an open horizon of specific, historically and culturally constituted, sociomaterial relations. Crucially, these relations are still power differentiated but in ways that can be recovered, as distributions located in specific configurations. Although the cyborg since Haraway suggests generative new forms of analysis, however, to realize that promise requires shifting out from its popular figuring as a singular, albeit hybrid, entity. The latter inherits a problem that characterizes any strategy centered on a heroic (even monstrous or marginalized) figure; that is, it obscures the presence of distributed sociomaterialities in more quotidian sites of everyday life. Along with the dramatic possibilities of the feminist cyborg, we need to recover the ways in which more familiar bodies and subjectivities are being formed through contemporary interweavings of nature and artifice, for better and worse.[18] Put another way, now that the cyborg figure has done its work of alerting us to the political effects, shifting

[18] This includes, for example, the Silicon Valley workers identified by Sandoval (1995), who "know the pain of the union of machine and bodily tissue" as they assemble the components of new objects within old regimes of racially and ethnically based difference. Relatedly, Jain (1999) considers the multiple ways in which prostheses are wounding at the same time that they are enabling. In contrast to the easy promise of bodily augmentation, she observes, the fit of bodies and artifacts is often less seamless and more painful than the trope of the cyborg would suggest. Jain (2006) takes legal contests over injury as a public and consequential site for the exploration of attributions of agency across the person–artifact boundary, within the wider dynamics of American commodity culture. Subject to Jain's insightful analysis, normative debates over things and their social consequences provide evidence for how the worlds that we inhabit are configured and by whom. This question is further elaborated by Schull (2005, in press),

boundaries, and transformative possibilities in human–machine mixings, it is time to get on with investigation of particular configurations and their consequences. How then might we locate conditions for action and possibilities for intervention in the specificities of more mundane sociomaterial assemblages?

DESIGN PRACTICES

Over the twenty years since the original publication of *Plans and Situated Actions*, new developments in professional practices of computer systems design have at least provided existence proofs of transformative possibilities. The emergence of increasingly distributed, networked computing during the late 1970s and early 1980s raised questions that clearly went beyond the limits of the human–machine interface narrowly construed to involve collective forms of computer use.[19] The turn to the social among computer scientists and information systems designers in the mid-1980s was accompanied by an intensification of interest among social researchers in the material grounds of sociality. Within ethnomethodology and conversation analysis, a growing awareness of the centrality of nonvocal activities (most obviously gaze and gesture) to the organization of face-to-face human interaction inspired a move toward the incorporation of materially based activity into the field of study. From Charles Goodwin's attention to the lighting of a cigarette in *Conversational Organization* (1981) to Goodwin and Goodwin's analyses of the interactional organization of eating and talk at a family dinner (1992), Heath's attention to the interactional enactment of patient pain (1986), and Schegloff's observations regarding the interactional effects of body "torque" (1998), interaction analysts increasingly recognized the interorganization of talk and other forms of embodied activity. Among

in her compelling account of the slippage between autonomy and automaticity in the case of human–machine couplings at the interface of video gambling machines.

19 The phrase Computer-Supported Cooperative Work (CSCW) was coined by Irene Greif, then on the faculty in Computer Science and Electrical Engineering at MIT, to convene a small invited workshop in 1984. This led to a series of still ongoing biannual conferences, as well as events held in alternate years under the title of the European CSCW or ECSCW conferences. CSCW is now an established subfield within professional networks of research and development engaged with the design of computer-based systems and devices. The CSCW conferences and journal have been the primary site for both programmatic and empirically based discussion between researchers in the computing and social sciences and the venue for a rich corpus of technical explorations and ethnographically informed investigations of technology-intensive sites of social action.

those of us immersed in ethnomethodology and newly engaged with the enterprise of computer systems design, absence of attention to the social and material organization of relevant forms of practice – from following instructions in the operation of photocopier to maintaining the order of traffic in the air – was an obvious site for generative intervention. The corpus of studies is by now extensive and comprises an established resource in the repertoire of design for technology-intensive forms of practice across a range of settings.

A central argument of these studies is that the nature and relevance of environment, objects, and actions are reflexively constituted through the ongoing activities of their habitation, engagement, and recognition. In the context of administering organizations operating across widely distributed locales, moreover, many of the relevant objects materialize technologies of coordination and control – procedural instructions, schedules, protocols, and the like – that prescribe courses of action designed to be reliably reproduced or available for comparative assessment. Relevant artifacts include, for example, flight progress control strips (Hughes, Randall, and Shapiro 1993), airline schedules (Goodwin and Goodwin 1996; Suchman 1993b), and railway timetables (Heath and Luff 1992). The politics of such artifacts (as for any technologies) include relations between the sites and interests within which coordinative artifacts are generated and those of their use. Like the "plan" that forms a focal object for this book, such technologies presuppose an open horizon of sociomaterial practices that inevitably exceed their representational grasp. At the same time, those practices reflexively constitute themselves as implementation of the actions prescribed. As I discussed at length in Chapter 11, the frequent presence of multiple, often contradictory, agendas of workplace auditing, on the one hand, and the work required to enact an orderliness within the work, on the other, lead to various forms of both breakdown and creative resistance. Design for such settings is therefore an inherently ethical project (Robertson 2002: 300).

Whereas Computer-Supported Cooperative Work directs the attention of researchers and systems designers to the sociality of computer use, a second, intersecting research community has taken up the challenge of a more radically conceived interference in existing arrangements of professional systems design. Inspired initially by pilot projects in the Nordic countries, involving codevelopment of information systems among organized workers and politically astute computer scientists, the project of participatory design entered the awareness of North

American researchers in the 1980s.[20] The by-now extensive body of research that has been conducted under the auspices of participatory design by no means conforms to a single orthodoxy.[21] The guiding commitment, however, is to rethinking critically the relations between practices of professional design and the conditions and possibilities of information systems in use. Central to this process is an attunement to the politics of design, that is, an orientation to the inevitable interrelations of agendas of technological change and (re-)distributions of labor with associated implications for both material and symbolic reward.

A common premise for both CSCW and participatory design, arising from their basis in empirical investigations of technologies-in-use across a range of settings, is that design – the configuration of artifacts – is not the exclusive province of professional practitioners. The necessity and creativity of ongoing practices of design-in-use has by now been extensively documented. Rather than a process that stops at the point of hand-off from production to consumption, design is as an ongoing process of (re-)production over time and across sites. Just what, then, is the role of the professional designer? Although in no way obviating the specific knowledges and material practices of the designer, the object of design must shift. Rather than fixed objects that prescribe their use, artifacts – particularly computationally based devices – comprise a medium or starting place elaborated in use. Rather than holding stable and separate the identities of "designer" and "user," the latter work as categories

[20] Invited by Irene Greif to act as Program Chair for CSCW 1988 and newly aware of activities in Scandinavia, I welcomed the opportunity to encourage this exchange through a series of papers presented at the second annual CSCW conference in 1988. A more dedicated conference was convened in 1990 under the auspices of Computer Professionals for Social Responsibility, with the title Participatory Design of Computer Systems or PDC (see Schuler and Namioka 1993), and these conferences have continued biannually since. For founding volumes in this area see Bjerknes, Ehn, and Kyng (1987), Ehn (1988), and for more recent collections see Greenbaum and Kyng (1991), Schuler and Namioka (1993).

[21] Various of the ideas and design practices characteristic of participatory design have by now made their way – more and less unscathed – into mainstream circulation under the rubric of user-centered design. For thoughtful introductions see Carroll (2000); Landauer (1995); Rogers, Sharp, and Preece (2002). At Xerox PARC during the 1990s I and my colleagues characterized our approach as one of practice-based codesign. Our aim in associating with these particular terms was not to stake out new terrain, but on the contrary to avoid the inexorable slide toward what Verran has named "hardening of the categories" that comes with the repetition (and initial captialization) of naming, particularly in the context of competitive R&D. Our intent was to maintain the provisionality and fluidity of our self-descriptions, while acknowledging our relations and indebtedness to an extended research community.

describing persons differently positioned, at different moments, and/or with different histories and future investments in projects of technology development (see Suchman 1999, 2002a, 2002b).

INHABITING THE INTERFACE

In her analysis of computer-based work, Susanne Bødker (1991) has discussed the shifting movement of the interface from object to connective medium. She observes that when unfamiliar, or at times of trouble, the interface itself becomes the work's object. At other times persons work, as she puts it, "through the interface," enacted as a transparent means of engagement with other objects of interest (for example, a text or an interchange with relevant others). As a case in point, we can consider the reflections of a civil engineer working at a CAD workstation (see also Henderson 1999; Suchman 2000). Although CAD might be held up as an exemplar of the abstract representation of concrete things, for the practicing engineer the story is more complex. Rather than stand in place of the specific locales – roadways, natural features, built environments, people, and politics – of a project, the CAD system connects the experienced engineer sitting at her worktable to those things, at the same time that they exceed the system's representational capacities. The engineer knows the project through a multiplicity of documents, discussions, extended excursions to the project site, embodied labors, and accountabilities: the textual, graphical, and symbolic inscriptions of the interface are read in relation to these heterogeneous forms of embodied knowing. Immersed in her work, the CAD interface becomes for the engineer a simulacrum of the site, not in the sense of a substitute for it but rather of a place in which to work with its own specific materialities, constraints, and possibilities. Like the symbiotic gesture described by Goodwin (2003), the CAD interface in use associates disparate elements both within and beyond its frame at the same time that those elements are essential to its intelligibility and efficacy.

Feminist film theorist Laura Marks describes what she calls "haptic visuality" as comprising "images that encourage a sympathy, intimacy and complicity between work and viewer" (2002: 3). She uses the term *work* in this context not with reference to a fetishized object resulting from cinematic practice but as an always only partially representable complex of social and material labors. Such works effect what Marks calls a "three-dimensional intimacy" among persons, images and their materiality, and the worlds to which the images connect. Those intimacies, in turn, dissolve the space between object and subject, evoking an embodied response that is more a form of inhabiting the cinematic work than a distanced appraisal of it. Central to such affective effects are the specific materialities of the medium. Marks observes that in the early days of cinema filmmakers demonstrated their fascination with the new medium's materialities as much as with its abstract representational power. Seen from this perspective, film is not a neutral conveyor of images, but rather the particular qualities of film stock are themselves an integral part of the imagery and imaginary created. It is not only the film's materialities, moreover, that this approach is aimed to recover but also "the rarely acknowledged workers who toil behind the scenes" (ibid.: 8). Viewing film is not then a matter of observer and image but an encounter among the efforts and effects of specifically situated persons and things.

Artist Heidi Tikka, in her work titled *Mother, Child*, offers an indicative case in point of a practice that plays with the multiplicity of materialities involved in so-called new or digital media. This work, which I had the opportunity to experience during its exhibition at the Art Gallery of Ontario in Toronto, Canada, in 2001, employs the shifting dynamics of installation, viewer–user, and onlookers, as well as the ambient environment of the exhibition space to invoke, and affectively evoke, an encounter between caregiver and infant. The piece does this not "in general" but always specifically: the caregiver is one particular visitor who enters the space of the installation and sits on a chair, and the infant is one particular infant (Tikka's son, recorded on digital video). A deliberate aspect of the piece is the heterogeneity of its forms: actual bodies and objects combine with projected images to comprise a hybrid of social and material elements. Together these elements create an interactive space characterized by a mix of predictability and contingency – a fragile stability – that affords the installation its affective kinship to the "real-world" encounter that it simulates. The three-dimensional image of a child that is projected – both technically and psychically – onto the soft cloth diaper that the viewer–user holds in her lap can be affected through her motions and orientation to it but dissolves as she stands and places the cloth back onto the chair. In this and other ways, the installation continually reminds us of, rather than conceals, its artifice. As Tikka herself comments, the piece is actually simpler (less reactive) in its composition than we experience it to be. The effects are created through the particular possibilities provided by an artful integration of persons, objects, spaces, fantasies, remembered experiences, and technologies to evoke and explore an emblematically human encounter but not to replicate it.

I would propose that in such projects the specific materialities of computing are under investigation, and reconfiguration, in forms that more radically challenge traditional imaginaries of the human than do the most ambitious projects in humanoid robotics. Central to this innovative approach is abandonment of the project of the "smart" machine endowed with capacities of recognition and autonomous action. Computational media artist Sha Xin Wei works with what he names "responsive media spaces" like the *Tgarden*, an installation populated by specially costumed participants instrumented with sensors, real-time tracking receivers, and media-synthesis generators. As Xin Wei describes it:

> The TGarden software tracks gesture rather than recognizes gesture, because at no place in the software is there a 'model' that codes the gesture ... The software does not infer what the player means by her gesture, it merely tracks the gesture and continuously synthesizes responses. So what we have done is to set aside entirely the problem of inferring human intent from behavior, or more generally from observables. Yet by providing and even thickening the sensuous response, we make fertile the substrate for agency. This approach remains agnostic as to whether movements are intentional; the responsive system simply does not need to know. (2002: 457)

More than conversation at the interface, it is creative assemblages like these that explore and elaborate the particular dynamic capacities that digital media afford and the ways that through them humans and machines can perform interesting new effects. Not only do these experiments promise innovations in our thinking about machines, but they open up as well the equally exciting prospect of alternate conceptualizations of what it means to be human. The person figured here is not an autonomous, rational actor but an unfolding, shifting biography of culturally and materially specific experiences, relations, and possibilities inflected by each next encounter – including the most normative and familiar – in uniquely particular ways.

Media scholar Chris Chesher (2004) has proposed a vocabulary of encounters with computer-based art that suggestively reworks information theoretic tropes at the interface. Although his proposal is applicable, I believe, to any example of human–computer interaction, Chesher starts from the premise that new media artists' noninstrumental applications of technology put the distinctiveness of computer-based forms into greater relief. From his consideration of new media art, Chesher proposes the concept of *avocation* to describe the arrangements and affordances through which persons are hailed to enter into a particular technological assemblage to become incorporated as integral actants in an associated form of sociomaterial agency. These include not only instrumental possibilities but also multiple and uncertain ways in which "new media art distracts and summons its users." *Invocation* involves those actions that define the terms of engagement written into the design script or discovered by the participating user, the calling up of events that effect changes to the assemblage. And finally, *evocation* describes the affective and effective material changes that result; transformations that in turn comprise the conditions of possibility for subsequent avocations. Together these terms articulate the distinctive dynamics of computing and the modes of engagement that it makes possible. The latter are characterized by what Chesher names a form of "managed indeterminacy," effected not only by databases and central processing units but crucially by "the peripherals that are in contact with materiality" that opens out from the boundaries of the machine narrowly construed. Offered in part as replacements for the more familiar terms *input*, *processing*, and *output*, these new "primitive technocultural formations" expand the space of interaction from the interface narrowly defined to the ambient environments and transformed and transformative subject–object relations that comprise the lived experience of technological practice.

Chesher's generative discussion of the distinguishing role of invocations in human–computer interactions provides a new basis on which to consider the forms of asymmetry that I described in *Plans and Situated Actions*. In particular, I identified as crucial what I characterized then as the machine's access to the activities of the user, limited specifically to those that changed its state. Working outside the bounds of AI's preoccupations with the agencies of the interactive machine, Chesher is less concerned with questions of human–machine symmetry than with the forms of invocation available and their evocative effects. For both analyses the question of invocation is central, but Chesher's framing helps to shift the focus from a preoccupation with whether the machine is like the human to a consideration of specific sociomaterial assemblages, their possibilities, and their consequences. New media artists, their works, and the persons whom the latter engage are configured together through these assemblages. Within that, difference becomes the basis for more than the repetition of relations of power, command and control, or obedient service.

EXPANDING FRAMES AND ACCOUNTABLE CUTS

The past twenty years of scholarship in the humanities and social sciences have provided new resources for thinking through the interface of humans and machines. Reappropriation of the cyborg, as a figure whose boundaries encompass intimate joinings of the organic and inorganic, has provided a means of analyzing myriad reformations of bodies and artifacts, actual and imagined. Expanded out from the singular figure of the human–machine hybrid, the cyborg metaphor dissolves into a field of complex sociomaterial assemblages, currently under study within the social and computing sciences. From close readings of encounters at the interface of person and machine, through extended historical and comparative analyses of technology-intensive, distributed worksites, these reconceptualizations have opened a generative wave of new scholarship and practice.

Methodologically, this view of the nature of sociomaterial research objects has two profound consequences. First, it demands attention to the question of frames, of the boundary work through which a given entity is delineated as such. Beginning with the premise that discrete units of analysis are not given but made, we need to ask how any object of analysis – human or nonhuman or combination of the two – is called out as separate from the more extended networks of which it is part.[22] This work of cutting the network is, I have argued, a foundational move in the creation of sociomaterial assemblages as objects of analysis or intervention. In the case of the robot, or autonomous machine more generally (as in the case of the individual human as well), this work takes the form of modes of representation that systematically foreground certain sites, bodies, and agencies while placing others offstage. As I suggested in Chapter 14, this spatial attenuation of the relevant field of agencies is accompanied by the staging of performances repeatable over time through accounts and demonstrations that have themselves been congealed into modes of immutable mobility. Our task as analysts is then to expand the frame, to metaphorically zoom out to a wider view that at once acknowledges the magic of the effects created while

[22] This is what Law terms a *method assemblage* (2004: 14).

explicating the hidden labors and unruly contingencies that exceed its bounds.

At the same time, a full analysis needs to locate these entities and the sites and moments of their efficacy in still more extended spatial and temporal relations. Encounters at the interface invariably take place in settings incorporating multiple other persons, artifacts, and ongoing activities, all of which variously infuse and inform their course. Questions of scale in the social sciences have traditionally been conceived either as a matter of counting – how many more of these units of analysis are there – or of reformulation in more general terms. An alternative is to approach scale not as a matter of one to many, little to big, or specific to general but rather of extension in time and space. How far our analysis extends in its historical specificity and reach, or in following out lines of connection from a particular object or site to others, is invariably a practical matter. That is, it is a matter of cutting the network, of drawing a line that is in every case enacted rather than given. The relatively arbitrary or principled character of the cut is a matter not of its alignment with some independently existing ontology but of our ability to articulate its basis and its implications.

These methodological questions are not privileged issues for the social sciences but an endogenous aspect of every site of sociomaterial configuration. From the designer who must delineate the boundaries of system and user(s) to the surgeon's body reconfigured by telemetric vision or the nurse enrolled in redesign of an operating theatre, matters of joining and separation of human and nonhuman are everyday affairs. However entrenched through repetition or provisionally held together, these relations are enacted. The task for critical practice is to resist restaging of stories about autonomous human actors and discrete technical objects in favor of an orientation to capacities for action comprised of specific configurations of persons and things. To see the interface this way requires a shift in our unit of analysis, both temporally and spatially. Temporally, understanding a given arrangement of humans and artifacts requires locating that configuration within social histories and individual biographies for both persons and things. And it requires locating it as well within an always more extended network of relations, arbitrarily – however purposefully – cut through practical, analytical, and/or political acts of boundary making.

My concern in this book has been with the specific material-discursive apparatuses through which contemporary relations of humans and machines are rendered intelligible and made real. Karen Barad proposes that "reality is sedimented out of the process of making the world intelligible through certain practices and not others" (1998: 105). Barad's agential realism reminds us that boundaries between humans and machines are not naturally given but constructed in particular historical ways and with particular social and material consequences. As Barad points out, boundaries are necessary for the creation of meaning and, for that very reason, are never innocent. Because the cuts implied in boundary making are always agentially positioned rather than naturally occurring, and because boundaries have real consequences, "accountability is mandatory" (ibid.: 187). The accountability involved is not, however, a matter of identifying authorship in any simple sense but rather a problem of understanding the effects of particular assemblages and assessing the distributions, for better and worse, that they perform. As Barad puts it: "We are responsible for the world in which we live not because it is an arbitrary construction of our choosing, but because it is sedimented out of particular practices that we have a role in shaping" (ibid.: 102).

It is on this understanding of boundary making that I would propose that the price of recognizing the agency of artifacts need not be the denial of our own. Now that agencies of things are well established, might we not bring the human out from behind the curtain, so to speak, without disenchantment? This requires, among other things, that we acknowledge the curtain's role. Agencies – and associated accountabilities – reside neither in us nor in our artifacts but in our intra-actions. The question, following Barad, is how to configure assemblages in such a way that we can intra-act responsibly and generatively with and through them. The legacy of twentieth-century technoscience posits autonomous agency as a primary apparatus for the identification of humanness and takes as a goal the reiteration of that apparatus in the project of configuring humanlike machines. Initiatives to develop a relational, performative account of sociomaterial phenomena indicate a different project. This project is based in recognition of the particularities of bodies and artifacts, of the cultural–historical practices through human–machine differences are (re-)iteratively drawn, and of the possibilities for and politics of redistribution across the human–machine boundary. Figured as interactions between humans and machines, the question has been whether the latter are best treated as objects or might one day successfully mimic the capacity of the autonomous human subject. The alternative perspective suggested here takes persons and machines as contingently stabilized through particular, more and less durable, arrangements whose reiteration and/or reconfiguration is the cultural and political project of design in which we are all continuously implicated. Responsibility on this view is met neither through control nor abdication but in ongoing practical, critical, and generative acts of engagement. The point in the end is not to assign agency either to persons or to things but to identify the materialization of subjects, objects, and the relations between them as an effect, more and less durable and contestable, of ongoing sociomaterial practices.